Assignment-02

Design and Implementation of a Multi-Step Prompt Workflow for
Automated Interview Preparation and Personalized Mock Interview Generation

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1. Introduction

Interview preparation often requires more than simply asking an AI model to provide a list of questions. A useful preparation resource should understand the candidate's role, experience level, skills, and target topics, then organize questions from basic concepts to more challenging interview situations. When all these activities are placed in one prompt, the response may contain repeated questions, unsuitable difficulty, missing areas, or inconsistent formatting.

This assignment presents a multi-step prompt workflow for automated interview preparation and personalized mock interview generation. The workflow separates the process into three connected stages: candidate and topic analysis, question generation, and quality review. The output of each stage is passed to the next stage, creating a controlled pipeline that can be checked at different points before the final preparation material is produced.

2. Objective

The main objective is to design and implement a reliable prompt workflow that converts a candidate profile and target role into structured, relevant, and personalized interview preparation material. The workflow specifically aims to:

• Identify the candidate level, target role, skills, and important interview areas before generating questions.

• Organize questions from fundamental concepts to application-based and challenging questions.

• Generate clear questions with suitable answers, examples, and explanations using predefined constraints.

• Keep the generated material aligned with the approved interview plan.

• Apply a mandatory review stage to detect repetition, irrelevant questions, missing topics, weak answers, and formatting problems.

A secondary objective is to keep every stage focused on one responsibility so that its output can be evaluated independently before continuing to the next stage.

3. Workflow Design

The proposed workflow treats interview preparation as a pipeline of three specialized stages. The first stage analyzes the candidate profile and creates an interview plan. The second stage transforms that plan into questions, answers, and practice material. The third stage acts as a quality-control gate and checks whether the draft satisfies the required relevance, difficulty, coverage, and formatting criteria.

Stage 1 – Planning: Defines the candidate profile, target role, skill areas, difficulty levels, and interview structure.

Stage 2 – Drafting: Generates questions, model answers, examples, and practice prompts using only the approved plan.

Stage 3 – Review: Audits the preparation material, identifies failures, and produces a corrected final version.

This separation of responsibilities makes the workflow easier to control and reduces the risk of producing content that is incomplete, repetitive, or unsuitable for the candidate.

4. Methodology

The workflow was developed by creating three chained prompts. Each subsequent prompt explicitly receives or uses the output from the preceding stage. This ensures that the final interview material is based on an approved plan rather than being generated independently at every step.

4.1 Step 1: Candidate Profile and Interview Plan

The first stage converts a candidate profile and target job role into a structured interview plan. It identifies the candidate level, role requirements, technical areas, behavioral areas, difficulty levels, and recommended sequence. This provides a stable foundation for question generation.

Prompt 1: Act as an expert interview coach. Analyze the candidate profile and target role provided below. Identify the candidate level, role requirements, key technical skills, behavioral areas, three to five preparation objectives, recommended question sequence, and suitable practice activities. Create a structured interview preparation outline. Do not write the final questions yet. Candidate Profile: [Insert Profile] Target Role: [Insert Role]

4.2 Step 2: Mock Interview Question Generation

The second stage uses the approved interview plan to create the actual preparation material. The model is instructed to stay within the planned areas and provide questions appropriate for the candidate level. Explicit constraints help maintain consistent difficulty, structure, tone, and coverage.

Prompt 2: Using only the approved interview plan generated in Step 1, create the mock interview preparation material. Generate technical and behavioral questions according to the planned sequence. Provide clear model answers, short explanations, and examples where appropriate. Maintain a professional and practical interview-coaching tone. Do not introduce major topics outside the approved plan. Follow the requested length and formatting constraints.

4.3 Step 3: Critique, Validation and Final Refinement

The third stage introduces a mandatory quality gate. Instead of immediately accepting the generated preparation material, the model evaluates it against a checklist. Any issue found during the audit is corrected before the final version is presented.

Prompt 3: Act as a strict senior interview coach. Review the preparation material against this checklist: (1) Are all preparation objectives addressed? (2) Are questions relevant to the target role and candidate level? (3) Is the question sequence logically organized? (4) Are there unnecessary repetitions or unrelated questions? (5) Are the model answers clear, accurate, and practical? (6) Does the output follow the requested length and format? First list the issues found. Then provide the corrected final interview preparation material.

5. Results and Discussion

The multi-step workflow provides a systematic approach to automated interview preparation. The planning stage helps define what should be tested and how the questions should be organized before detailed content is generated. As a result, the drafting stage has a clear structure to follow and is less likely to omit important interview areas.

The second advantage is controlled content generation. Because the drafting prompt refers to the approved interview plan and specifies candidate level, role relevance, tone, organization, and formatting requirements, the generated material can remain focused on the intended interview. The process also makes it easier to modify the plan before generating the complete question set.

The final critique stage provides an additional safeguard. It can identify issues such as missing topics, unclear answers, repeated questions, irrelevant content, unsuitable difficulty, or violations of formatting rules. Correcting these issues before delivery makes the final preparation material more consistent and useful.

Overall, the workflow demonstrates that dividing a complex interview-preparation task into smaller prompt-driven stages can improve controllability and make the generation process easier to evaluate. Each stage creates an intermediate artifact that can serve as a checkpoint for the next stage.

6. Conclusion

The design and implementation of a multi-step prompt workflow shows how prompt engineering can be applied to automated interview preparation and personalized mock interview generation. Instead of depending on one overloaded prompt, the proposed approach uses planning, question generation, and critique as three cooperating stages.

The workflow creates clear checkpoints for validating the candidate profile, interview structure, generated questions, model answers, and final quality. It can be adapted to different roles and experience levels by changing the target role, skills, objectives, difficulty, examples, and formatting requirements.

Therefore, a multi-step prompt workflow provides a practical and repeatable method for producing structured and personalized interview preparation resources with improved consistency and controllability.